Who Touched My Sentence? Inference for AI-Edited Documents in the Sparse and Weak Regime

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Abstract

Much AI-assisted writing is drafted by a person and then edited by an AI model, so a document can contain AI-edited sentences among purely human ones. The editing can be dense or sparse, down to a few changed words in a single sentence. We treat each edited sentence as a sparse signal and test whole documents with Higher Criticism (HC) applied to per-sentence conformal p-values. To train sentence detectors, we collect 313k human sentences from eight domains, written in 2020 or earlier, and have six current LLMs edit every train, validation and test sentence under two prompts. Five editors are used for training and the sixth is held out. Our ensemble of four fine-tuned detectors flags 38.6% of edited test sentences at a 1% false-positive rate (36.2% for the held-out editor). The strongest released detector overall that we evaluated, EditLens, flags 8.7% (it was trained on passages of 75 to 799 words, so single sentences are outside its training range), and zero-shot detectors flag 1.5 to 2.1%. On test documents of at least 16 sentences, HC keeps its false-positive rate at the 5% level once its thresholds account for correlation between sentences of the same document. It flags 92% of documents in which 65% of the sentences are edited and 22% of documents in which a single sentence is edited. EditLens applied to the whole document, its intended use, flags 53% and 6% of the same documents. A second test separates human-drafted, AI-edited documents from fully AI-generated ones. It labels 96% of human documents and 93% of fully generated documents correctly, and 87% of densely edited documents as AI-edited. The detectors are not retrained for this test. They were trained only to separate human sentences from AI-edited ones, yet their scores rise with the amount of editing and they flag nearly all fully generated documents. Finally, we study language models trained on AI-edited text, from small models trained from scratch to continued pretraining of Pythia-6.9B. Training on AI-edited text makes a model prefer the AI-edited version of a sentence over its human original. When only some sentences are edited, as in real data, this shift barely changes perplexity on held-out human text, so perplexity alone would not reveal it. Down-weighting edited sentences by how much they were edited removes about a tenth of the shift, and dropping them removes nearly all of it at a cost in perplexity. On seven public benchmarks, at the same 5% false-alarm rate, the ensemble beats EditLens and MELD where edits are light or sparse, and trails them on open-ended rewrites and on fully generated text.

1 Introduction

Much AI-assisted writing, such as student essays, peer reviews, emails and scientific papers, comes from revision between a person and an AI model. The person drafts, the model edits, and the two may alternate several times. The result is a document in which an unknown number of sentences, at unknown positions, were edited by AI, each to an unknown degree. Detectors for fully AI-generated text are well developed and widely used. Edited text is harder: detectors flag even lightly polished human text and struggle to tell degrees of AI involvement apart [1], and the edits are often confined to a few sentences of a long document. Finding a few weak signals among many null observations is the setting of sparse inference, and we use its tools.

A document of n sentences gives n hypotheses, one per sentence, where $H_{0,i}$ states that sentence i is human-written. A sentence-level detector turns each sentence into a p-value that is exactly uniform when the sentence is human. We then test the global null H_0, that every sentence of the document is human, with Higher Criticism (HC) [2, 3] and related tests. We also estimate which sentences were edited and what fraction of the document they make up.

Two quantities describe an edited document in our experiments. The sparsity $\theta \in (0, 1)$ sets the number of edited sentences, $m = \text{round}(n^{1-\theta})$. At $\theta = 0.1$ most sentences are edited, and at $\theta = 0.9$ a 16-sentence document has one edited sentence. The edit rate r of a sentence is the fraction of its words that the AI changed (defined in Section 2.2). We compare our results with the Gaussian rare/weak model [4, 2], in which a human sentence has $z \sim N(0, 1)$ and an edited sentence has $z \sim N(\sqrt{2r^* \log n}, 1)$. Here r^* is a signal strength on the z scale and should not be confused with the edit rate r. In this model $\theta > 1/2$ is the sparse regime, where tests that sum evidence over all sentences lose power and HC remains effective.

Our contributions are as follows.

- A paired corpus of 313,249 human sentences from eight sources, split into train, val, calib and test sets by document. Every train, val and test sentence is edited by six LLMs under two prompts, about 2.1M accepted edits in total.
- Sentence detectors fine-tuned on these edits and their ensemble, compared with four released detectors and six zero-shot ones on our corpus and with three released detectors on seven third-party benchmarks.
- A document-level pipeline with exact per-sentence conformal p-values, global tests whose thresholds account for correlation between sentences of the same document, and estimates of which sentences and what fraction were edited. We evaluate it over the (θ, r^*) plane against the rare/weak theory.
- Results on separating AI-edited from fully generated documents, and on how training on AI-edited text changes a language model.

2 Setup

Figure 1 shows the pipeline in three parts. Part (a) builds the data, part (b) turns one sentence into a p-value, and part (c) tests one document.

2.1 Human corpus

Eight sources cover web pages, scientific papers, school essays, essays by English learners, news, peer reviews, emails and stories (Table 1). They are FineWeb-Edu [5], peS2o [6], PERSUADE 2.0 [7], ELLIPSE [8], XSum [9], OpenReview [10], Enron emails via AESLC [11, 12] and WritingPrompts [13]. We keep only text written in 2020 or earlier, so that it predates current LLM writing assistants. For sources with document dates we drop documents dated after 2020. The other sources contain only texts written before 2021 by the way they were collected.

For each source we fill four splits with whole documents until they hold about 15,000 train, 1,500 val, 15,000 calib and 7,500 test sentences, so no document appears in two splits. Calib sentences are never edited, because they serve as the human reference for p-values (Section 2.4). We split documents into sentences with Qwen3.5-122B [14] and accept its output only if the returned sentences, joined together, reproduce the input text apart from whitespace (quote styles may differ and hyphens at line breaks may be dropped). Otherwise we use SaT [15].

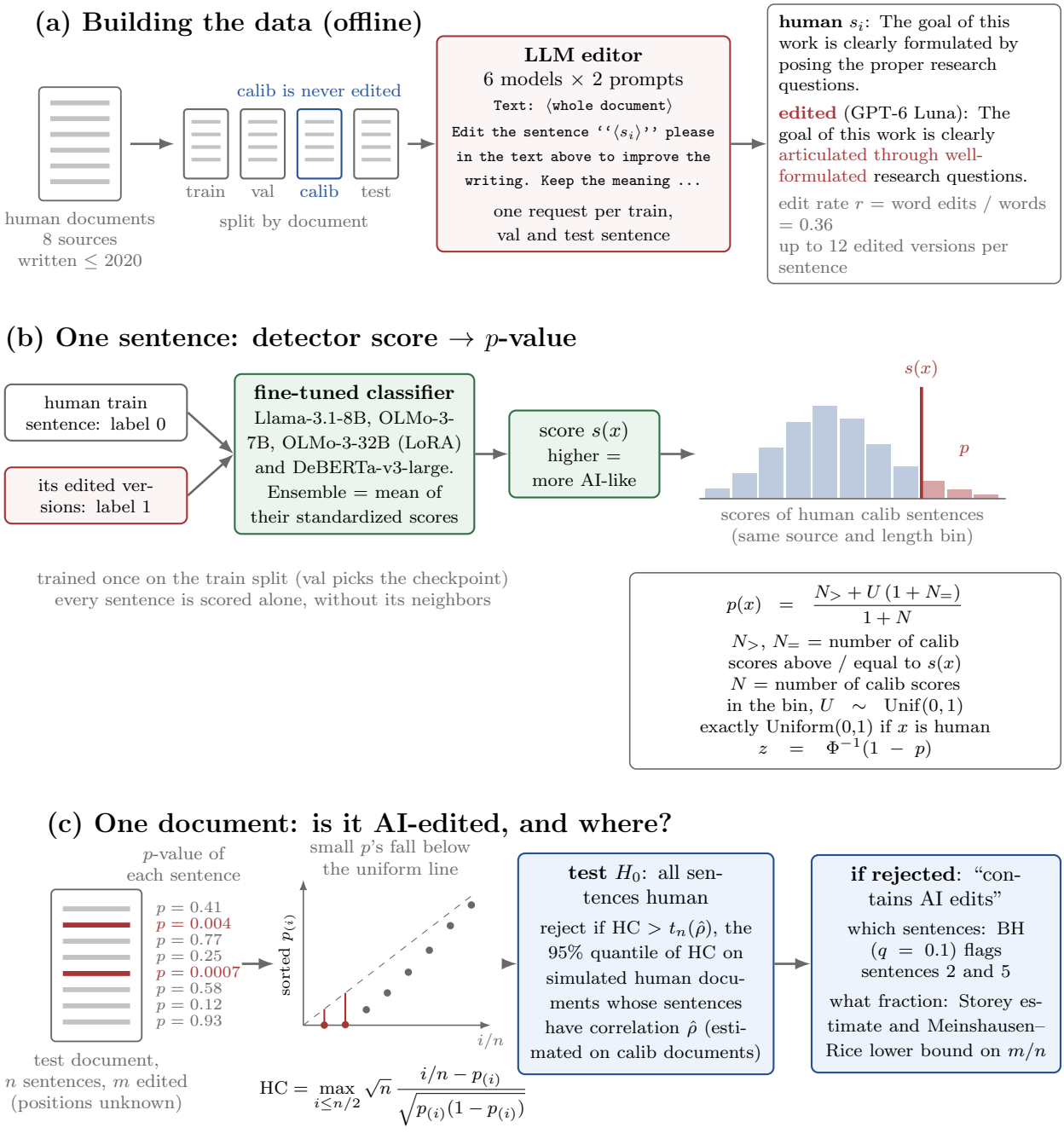


Figure 1. Pipeline. (a) Human documents written in 2020 or earlier are split by document into train, val, calib and test sets. LLMs edit every train, val and test sentence one at a time while seeing the whole document. As mentioned, calib documents are never edited. (b) A classifier trained to separate human train sentences from their edited versions scores each sentence on its own. The score becomes an exact p-value by ranking it among the scores of human calib sentences of the same source and length (we also test pooling sources, and separating calib only by length). (c) The n sentence p-values of a test document are combined by Higher Criticism (we also report min-p, HC$^+$ and Stouffer's test), with a threshold that allows for correlation between sentences of one document. Benjamini–Hochberg (BH) then locates the edits. Evaluation uses the 1,301 test documents with $n \geq 16$. Each document is edited by every editor and prompt at $m = \text{round}(n^{1-\theta})$ random positions for $\theta \in \{0.1, \ldots, 0.9\}$, and is also kept unedited to measure the type I error.

Source	Documents				Median sentences/doc	Sentences	Test docs $n \geq 16$
	train	val	calib	test			
FineWeb-Edu	386	31	364	218	27	39,412	149
peS2o	74	11	79	38	169	39,460	38
PERSUADE 2.0	713	74	690	349	20	39,034	241
ELLIPSE	784	81	787	376	18	39,027	231
XSum	828	69	779	404	15	39,046	203
OpenReview	681	73	652	344	19	39,081	207
Enron	1,885	181	1,880	1,010	5	39,067	77
WritingPrompts	305	32	301	175	40	39,122	155
Total	5,656	552	5,532	2,914	14	313,249	1,301

Table 1. Corpus. Test documents with at least 16 sentences form the document-level test set.

2.2 AI edits

Each train, val and test sentence is edited by six models, once under each of two prompts, so it has up to 12 edited versions. Five models are training editors, namely gpt-oss-120b [16], Gemma 4 31B [17], GPT-6 Luna [18], Claude Haiku 5.5 [19] and GLM-5.3 Flash [20]. The sixth, Qwen3.5-122B-A10B [14], is held out, and its edits are never used to train or select any of our detectors, to detect extrapolation to new model families. We do not believe the sentence splits being done by Qwen cause any problems to this end. The editor sees the whole original document and is asked to edit one sentence. The "edit" prompt reads *Edit the sentence "..." in the text above to improve the writing. Keep the meaning and about the same length, and keep it one sentence.* The "light" prompt reads *Lightly edit ..., changing only what needs it.*

We reject an output if it is empty, unchanged, splits or merges sentences, alters an anonymization placeholder, or is more than twice as long as the original. A rejected output is requested again, and a request that still fails after three attempts of sampling leaves the sentence without an edit from that editor and prompt (Table 2). The edit rate r of a sentence is the word-level Levenshtein distance between the original and the edit, divided by the number of words in the original. It can exceed 1 when the edit is longer than the original. The median r ranges from 0.19 (GLM-5.3 Flash, light prompt) to 0.50 (gpt-oss-120b).

Editor (prompt)	Role	Accepted edits			Rejected (%)	Edit rate r	
		train	val	test		median	mean
gpt-oss-120b	train	112,462	11,552	56,666	6.2	0.50	0.53
gpt-oss-120b (light)	train	111,618	11,496	56,257	6.8	0.36	0.40
gemma-4-31b	train	113,241	11,615	57,134	5.5	0.47	0.51
gemma-4-31b (light)	train	107,895	11,085	54,434	9.9	0.23	0.29
gpt-6-luna	train	110,629	11,343	55,808	7.7	0.38	0.43
gpt-6-luna (light)	train	106,058	10,941	53,542	11.4	0.23	0.28
claude-haiku-5.5	train	105,266	10,717	53,094	12.2	0.45	0.51
claude-haiku-5.5 (light)	train	92,471	9,450	46,447	22.9	0.20	0.26
glm-5.3-flash	train	113,382	11,622	57,263	5.3	0.33	0.40
glm-5.3-flash (light)	train	108,691	11,177	54,848	9.2	0.19	0.25
qwen3.5-122b	held out	110,397	11,257	55,637	7.9	0.39	0.43
qwen3.5-122b (light)	held out	105,619	10,813	53,157	11.9	0.25	0.29

Table 2. Editors and prompts. Rejected is the share of requested edits still rejected after three attempts. Edit rates are computed on the test split.

2.3 Sentence detectors

Fine-tuned detectors. We fine-tune four classifiers to separate human train sentences (label 0) from their edited versions by the five training editors (label 1). The training set has about 120k human and 1.08M edited sentences, and a class-weighted binary cross-entropy loss offsets the imbalance. Three classifiers are decoders with LoRA adapters [21] and a classification head on the last token, Llama-3.1-8B [22], OLMo-3-7B and OLMo-3-32B [23]. The fourth is a fully fine-tuned DeBERTa-v3-large [24]. These LLMs were chosen to not have an overlap with the data generation families. Each sentence is scored on its own, without its neighbors. A human sentence's score therefore does not depend on whether its neighbors were edited, so its p-value stays exactly uniform in any document.

We chose these models in a search over 29 configurations, selecting by the true-positive rate (TPR) at a 1% false-positive rate (FPR) on val sentences of the training editors. We used this criterion because the smallest p-values are what drive HC and Bonferroni. Fine-tuned models clearly beat frozen embedding models with a logistic-regression head (TPR of 0.30–0.32 vs. at most 0.17 in the search), and after full training the four classifiers reach a val TPR of 0.34–0.37. Appendix A gives the search, the training runs and the hyperparameters. The **ensemble** standardizes each classifier's score by its mean and standard deviation on human val sentences and averages the four standardized scores. Nothing is fit on calib or test data.

Released detectors. We compare with four released detectors, EditLens [25] (Llama-3.2-3B with LoRA and a 4-bucket head for the amount of AI editing, scored as the expected bucket), desklib v1.01 [26] (DeBERTa-v3-large), GPTTrace v8s [27] (Ettin-1B) and GPTTrace Ettin-150M [28]. Each is run as its authors released it, and for EditLens we checked that our setup reproduces Pangram's published scores (Appendix B). EditLens was trained only on passages of 75 to 799 words, so single sentences are outside its training range. We therefore apply each released detector in two ways. The first scores every sentence alone and feeds the scores into the same p-value pipeline as our detectors. The second scores the whole document (truncated to the model's input limit) and ranks that score among the scores of calib human documents of the same source, which gives a document-level conformal p-value.

Zero-shot detectors. We use Fast-DetectGPT [29] (analytic version), Binoculars [30] and the mean token log-likelihood, each computed with two model families, Falcon-7B [31] and Llama-3-8B [22]. Methods that need two models use the base model together with its Instruct version.

2.4 From scores to p-values

Scores are oriented so that higher scores are associated with more AI-like text. For a test sentence from source k with score s, let $\mathcal{N}_k$ be the scores of the calib human sentences from source k that fall in the same word-count bin (10 bins at the deciles of sentence length). The randomized conformal p-value is

$$p = \frac{\#\{s' \in \mathcal{N}_k : s' > s\} + U\,(1 + \#\{s' \in \mathcal{N}_k : s' = s\})}{1 + |\mathcal{N}_k|}, \qquad U \sim \mathrm{Unif}(0,1).$$

It is distributed as $\mathrm{Unif}(0,1)$ when the test sentence is human and exchangeable with the calib sentences of its source and length bin [32, 33]. We also use $z = \Phi^{-1}(1-p)$, which is large and positive for sentences that look AI-edited. In the results, the TPR at FPR α is the share of edited test sentences with $p < \alpha$, and we report next to it the realized FPR, the share of human test sentences with $p < \alpha$.

2.5 Documents, global tests and estimation

Edited test documents. The document-level test set consists of the 1,301 test documents with at least 16 sentences, each used whole. For each of the 12 editor and prompt pairs and each $\theta \in \{0.1, 0.2, \ldots, 0.9\}$, we draw $m = \mathrm{round}(n^{1-\theta})$ sentence positions uniformly at random and replace those sentences by their edited versions. A drawn position whose edits were rejected stays human, so some documents have fewer than m edits with roughly 10% of drawn positions staying human. The 1,301 unedited documents are used to estimate the type I error.

Global tests. We use four tests at level $\alpha = 0.05$. The main one is HC with $\alpha_0 = 1/2$,

$$\mathrm{HC}_n = \max_{1 \le i \le n/2} \sqrt{n} \, \frac{i/n - p_{(i)}}{\sqrt{p_{(i)}(1 - p_{(i)})}},$$

where $p_{(1)} \le \cdots \le p_{(n)}$ are the sorted sentence p-values. HC^+ takes the same maximum over $p_{(i)} \ge 1/n$ only [2]. Min-p uses the smallest p-value [34], and Stouffer's test uses $\sum_i z_i/\sqrt{n}$ [35].

Correlation within a document. Sentences of one document share an author and a topic, so their z values are positively correlated even when every sentence is human. Thresholds derived for independent p-values are then too low, and HC's type I error becomes 0.065–0.104 across all detectors, and 0.083–0.095 for ours, instead of 0.05 (Appendix, Table 15). We model the correlation as equicorrelation, $z_j = \sqrt{\rho}\, a + \sqrt{1-\rho}\, e_j$ with a and e_j independent $N(0,1)$. We estimate ρ for each source and detector on calib documents and set each test's critical value at every n by simulating this null (Appendix C). The estimates $\hat{\rho}$ range from 0.06 to 0.32 for our detectors. In our experiments, this performs better than PC adjusted HC except for the sparsest of documents, $\theta \ge 0.7$.

Locating edits and estimating the edited fraction. We estimate which sentences were edited with Benjamini–Hochberg at $q = 0.1$ [36] and, for comparison, with Bonferroni at level α. We score both by the Hamming error divided by m, that is, the number of missed plus falsely flagged sentences over m, and by how often they recover the edited set exactly. We estimate the edited fraction $\varepsilon = m/n$ with Storey's estimator at $\lambda = 1/2$ [37] and give a 95% lower bound with the Meinshausen–Rice method using the HC bounding function [38].

3 Results

3.1 Sentence level

At a 1% FPR, our ensemble detects 38.6% of the edited test sentences from the training editors and 36.2% of those from the held-out Qwen editor (Table 3). Each of the four classifiers alone detects 30–38%. The best released detectors reach 8.7% (EditLens) and 9.4% (GPTTrace v8s) on the training editors. The zero-shot detectors reach 1.5–2.1%, barely above the 1% that a detector with no signal would reach. Conformal calibration basically holds for every detector.

	Training editors			Qwen (held out)			Human: FPR		
Detector	0.1%	1%	5%	0.1%	1%	5%	0.1%	1%	5%
Ensemble	0.199	0.386	0.612	0.147	0.362	0.607	0.0011	0.0105	0.0495
OLMo-3-32B	0.193	0.377	0.599	0.140	0.352	0.596	0.0012	0.0111	0.0499
Llama-3.1-8B	0.169	0.341	0.571	0.103	0.295	0.546	0.0008	0.0102	0.0504
OLMo-3-7B	0.162	0.331	0.562	0.108	0.302	0.558	0.0010	0.0102	0.0496
DeBERTa-v3-large	0.170	0.341	0.573	0.121	0.316	0.565	0.0009	0.0101	0.0490
EditLens (3B)	0.018	0.087	0.243	0.022	0.098	0.251	0.0013	0.0108	0.0513
GPTTrace v8s (1B)	0.057	0.094	0.180	0.011	0.047	0.141	0.0009	0.0100	0.0504
desklib v1.01	0.002	0.022	0.083	0.003	0.024	0.086	0.0008	0.0102	0.0493
GPTTrace Ettin-150M	0.003	0.019	0.076	0.004	0.020	0.079	0.0010	0.0097	0.0503
Fast-DetectGPT (Llama-3-8B)	0.002	0.018	0.083	0.002	0.015	0.073	0.0014	0.0106	0.0514
Fast-DetectGPT (Falcon-7B)	0.002	0.015	0.068	0.001	0.012	0.058	0.0011	0.0095	0.0478
Log-likelihood (Llama-3-8B)	0.001	0.017	0.090	0.001	0.014	0.077	0.0009	0.0099	0.0490
Log-likelihood (Falcon-7B)	0.001	0.017	0.088	0.001	0.014	0.075	0.0008	0.0091	0.0471
Binoculars (Llama-3-8B)	0.003	0.021	0.088	0.002	0.018	0.078	0.0011	0.0106	0.0519
Binoculars (Falcon-7B)	0.003	0.017	0.074	0.002	0.014	0.063	0.0010	0.0096	0.0481

Table 3. Sentence level, test split. TPR is the share of edited sentences with p below the FPR level, shown for the training editors (545,493 edited sentences) and the held-out Qwen editor (108,794). FPR is the realized rate on 60,483 human test sentences. Blocks are our detectors (ensemble and its four members), released detectors and zero-shot detectors.

Detection rates vary depending on the editor used on the sentence (Table 4). The ensemble catches 66% of the edits by gpt-oss-120b but only 19% of the light edits by GLM-5.3 Flash. Within an editor it also, expectedly, depends on how much was changed (Table 5). The ensemble catches 19–23% of sentences with $r \leq 0.1$ and 53% of sentences with $0.5 < r \leq 1$. The held-out Qwen editor falls in the middle of the training editors, so there is no sign that the detectors overfit to the editors they were trained on. GPTTrace v8s is the only released detector with a clear editor preference, as it detects 25% of the edits by gpt-oss-120b at a 0.1% FPR and at most 1.7% of any other editor's. Detection also varies by domain (Appendix, Table 14), from 22% on FineWeb-Edu to 61% on ELLIPSE learner essays, where editors fix many errors.

Editor (prompt)	Ensemble	OLMo-3-32B	Llama-3.1-8B	OLMo-3-7B	DeBERTa-v3-large	EditLens (3B)	GPTTrace v8s (1B)
gpt-oss-120b	0.661	0.651	0.604	0.598	0.613	0.133	0.308
gpt-oss-120b (light)	0.576	0.568	0.537	0.524	0.535	0.095	0.286
gemma-4-31b	0.449	0.436	0.358	0.379	0.392	0.126	0.065
gemma-4-31b (light)	0.258	0.246	0.217	0.216	0.218	0.061	0.030
gpt-6-luna	0.438	0.431	0.405	0.363	0.362	0.096	0.057
gpt-6-luna (light)	0.313	0.307	0.296	0.257	0.259	0.067	0.042
claude-haiku-5.5	0.417	0.403	0.348	0.342	0.365	0.090	0.040
claude-haiku-5.5 (light)	0.242	0.233	0.211	0.202	0.213	0.061	0.025
glm-5.3-flash	0.283	0.273	0.234	0.235	0.250	0.075	0.035
glm-5.3-flash (light)	0.191	0.183	0.169	0.161	0.168	0.054	0.026
qwen3.5-122b	0.425	0.415	0.341	0.354	0.373	0.118	0.058
qwen3.5-122b (light)	0.295	0.286	0.248	0.247	0.255	0.076	0.036

Table 4. TPR at 1% FPR by editor and prompt, test split. Qwen rows are the held-out editor.

Edit rate r	Sentences	Ensemble	OLMo-3-32B	EditLens (3B)
≤ 0.05	12,391	0.225	0.214	0.035
0.05–0.1	53,355	0.190	0.183	0.035
0.1–0.2	121,871	0.248	0.242	0.046
0.2–0.3	110,141	0.337	0.329	0.067
0.3–0.5	196,830	0.443	0.432	0.097
0.5–1	148,440	0.533	0.521	0.153
> 1	11,259	0.319	0.305	0.075

Table 5. TPR at 1% FPR by edit rate r, all editors pooled. The bin $r > 1$ holds edits whose word-level edit distance exceeds the original's word count.

3.2 Document level

On the 1301 unedited human test documents, every detector's false-positive rate stays close to the nominal 5% once the thresholds account for correlation between sentences of the same document (Appendix, Table 15). For HC it ranges from 0.041 to 0.058 across detectors, and over all four tests the largest is 0.071, for min-p of OLMo-3-32B.

Table 6 summarizes HC, and Appendix Table 16 gives every θ. The ensemble rejects 92% of edited documents at $\theta = 0.1$, where on average 22.5 sentences, or 65% of the document, are edited. It rejects 59% at $\theta = 0.5$ and 22% at $\theta = 0.9$, where a 16-sentence document has a single edited sentence. Documents edited by the held-out Qwen editor are detected at nearly the same rates. The released detectors are far behind on both of their combined sentence scores by HC and their score on the whole document. The strongest of them, EditLens, rejects 46% of documents at $\theta = 0.1$ when its sentence scores are combined by HC and 53% when it scores the whole document, and at most 18% from $\theta = 0.5$ on. The zero-shot detectors reject at most 13% at $\theta = 0.1$. For EditLens, combining sentence scores with our HC pipeline beats scoring the whole document for every $\theta \geq 0.3$, so the sentence-level approach helps a released detector too once edits are spread thinly. This is especially striking, as EditLens was not trained to be a sentence level detector. Thus, we believe our pipeline shows promise for repurposing existing document level detectors as sentence level detectors and improving their downstream performance. However, this depends on the type of data on the third-party benchmarks, whose texts are more densely edited, whole-document scoring is better for EditLens (EditLens gets 0.46 per sentence vs 0.59 whole on OpAI-Bench). On our own corpus the ensemble is strongest on dense edits, and on third-party data it trails EditLens only where much of each text is AI-written (the EditLens test set, AITDNA and RAID).

By editor and by test. By editor and prompt (Table 7), the ensemble's HC power at $\theta = 0.1$ ranges from 0.76 (Claude Haiku 5.5, light prompt) to 0.999 (gpt-oss-120b), and at $\theta = 0.9$ from 0.10 to 0.45. Min-p is more powerful than HC when edits are sparsest ($\theta \geq 0.7$) and less powerful when they are dense, which falls within our expectations. Stouffer's test is below HC everywhere and closest at $\theta = 0.1$ (0.87 vs. 0.92). Appendix Tables 17 and 18 give the full results.

Comparison with the rare/weak theory. Figure 2 places each (document, editor, θ) triple at its effective signal strength $r^*_{\text{eff}} = \max(\mu_{\text{eff}}, 0)^2/(2 \log n)$, where μ_{eff} is the mean z of the document's edited sentences. This converts the observed z values to the r^* scale of the Gaussian model. The ensemble's empirical power follows the power that the Gaussian model predicts at the same n.

Robustness of the thresholds. The simulated thresholds assume equicorrelation. As a check that needs no dependence model, we also calibrated each test against the same statistic computed on calib human documents of the same source and length. For HC this changes almost nothing (type I error 0.043–0.046, power within 0.03), so the equicorrelation model is adequate for HC. Variants of HC that first remove the component shared by all sentences of a document (PC-adjusted and innovated HC) gain power only when edits are sparsest. With all tests calibrated on calib documents, the ensemble at $\theta = 0.9$ reaches 0.28 with these variants against 0.21 with HC. They lose heavily when edits are dense (0.23 and 0.44 vs. 0.91 at

Detector	Type I ($\alpha = 0.05$)	Detection: HC rejection rate ↑			Recovery: BH Hamming error/m ↓		
		$\theta = 0.1$	$\theta = 0.5$	$\theta = 0.9$	$\theta = 0.1$	$\theta = 0.5$	$\theta = 0.9$
Ensemble	0.045	0.92/0.93	0.59/0.56	0.22/0.19	0.46/0.46	0.68/0.70	0.87/0.89
OLMo-3-32B	0.054	0.91/0.93	0.58/0.55	0.22/0.19	0.47/0.47	0.69/0.71	0.89/0.90
Llama-3.1-8B	0.048	0.89/0.89	0.53/0.47	0.20/0.16	0.52/0.54	0.73/0.78	0.88/0.92
OLMo-3-7B	0.048	0.89/0.91	0.52/0.49	0.19/0.16	0.52/0.52	0.73/0.75	0.87/0.89
DeBERTa-v3-large	0.041	0.90/0.92	0.55/0.53	0.21/0.18	0.50/0.50	0.72/0.73	0.88/0.91
EditLens (3B)	0.058	0.46/0.50	0.17/0.18	0.08/0.08	0.87/0.86	0.97/0.96	1.07/1.07
GPTTrace v8s (1B)	0.049	0.35/0.29	0.19/0.11	0.09/0.06	0.92/0.96	0.97/1.00	1.01/1.04
desklib v1.01	0.053	0.14/0.14	0.07/0.08	0.06/0.06	0.99/0.99	1.02/1.02	1.08/1.07
GPTTrace Ettin-150M	0.058	0.13/0.14	0.07/0.08	0.06/0.06	0.99/0.99	1.02/1.03	1.07/1.07
Fast-DetectGPT (Llama-3-8B)	0.054	0.11/0.09	0.07/0.06	0.06/0.06	0.99/0.99	1.01/1.02	1.05/1.05
Fast-DetectGPT (Falcon-7B)	0.051	0.09/0.07	0.06/0.06	0.05/0.05	0.99/0.99	1.02/1.02	1.05/1.06
Log-likelihood (Llama-3-8B)	0.046	0.11/0.08	0.05/0.05	0.05/0.05	0.99/1.00	1.03/1.03	1.09/1.08
Log-likelihood (Falcon-7B)	0.044	0.10/0.07	0.05/0.05	0.04/0.04	0.99/0.99	1.02/1.02	1.08/1.08
Binoculars (Llama-3-8B)	0.058	0.13/0.10	0.07/0.07	0.06/0.06	0.99/0.99	1.01/1.02	1.06/1.06
Binoculars (Falcon-7B)	0.050	0.10/0.07	0.06/0.06	0.05/0.05	0.99/0.99	1.01/1.01	1.04/1.04
EditLens (3B), whole doc.	0.050	0.53/0.56	0.13/0.11	0.06/0.05	–	–	–
GPTTrace v8s (1B), whole doc.	0.055	0.39/0.26	0.13/0.07	0.07/0.06	–	–	–
desklib v1.01, whole doc.	0.051	0.35/0.34	0.11/0.09	0.06/0.05	–	–	–
GPTTrace Ettin-150M, whole doc.	0.055	0.09/0.08	0.05/0.05	0.05/0.05	–	–	–

Table 6. Document level, whole test documents with $n \geq 16$ and $m = \text{round}(n^{1-\theta})$ edited sentences. Each cell shows training editors / held-out Qwen editor. Detection is the HC rejection rate at $\alpha = 0.05$. Recovery applies BH at $q = 0.1$ to the document's sentence p-values and reports (missed + falsely flagged sentences)/m. The four blocks are our detectors, released detectors with sentence scores combined by HC, zero-shot detectors, and released detectors scoring the whole document directly, i.e their normal pipeline (document-level conformal test, which gives no sentence-level recovery).

$\theta = 0.1$), because removing the document's mean z also removes much of the edits' signal. Appendix D gives details.

Unknown source. Our null distributions are per source, which assumes the source is known. If each sentence is instead ranked against calib sentences of all sources, the overall sentence FPR stays at 1%, but the FPR per source ranges from 0.4% to 2.2% for the ensemble, and human FineWeb-Edu documents are flagged 13–17% of the time (Appendix E).

Editor (prompt)	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
gpt-oss-120b	1.00	0.99	0.93	0.76	0.45
gpt-oss-120b (light)	1.00	0.97	0.89	0.68	0.39
gemma-4-31b	0.98	0.90	0.70	0.48	0.24
gemma-4-31b (light)	0.87	0.65	0.39	0.25	0.13
gpt-6-luna	0.98	0.89	0.71	0.51	0.27
gpt-6-luna (light)	0.93	0.75	0.53	0.34	0.18
claude-haiku-5.5	0.97	0.84	0.62	0.40	0.20
claude-haiku-5.5 (light)	0.76	0.51	0.33	0.22	0.12
glm-5.3-flash	0.92	0.74	0.47	0.29	0.16
glm-5.3-flash (light)	0.78	0.52	0.33	0.21	0.10
qwen3.5-122b	0.97	0.86	0.67	0.46	0.24
qwen3.5-122b (light)	0.90	0.70	0.46	0.29	0.14
All editors	0.92	0.78	0.59	0.41	0.22

Table 7. Ensemble HC rejection rate by editor and prompt, whole test documents.

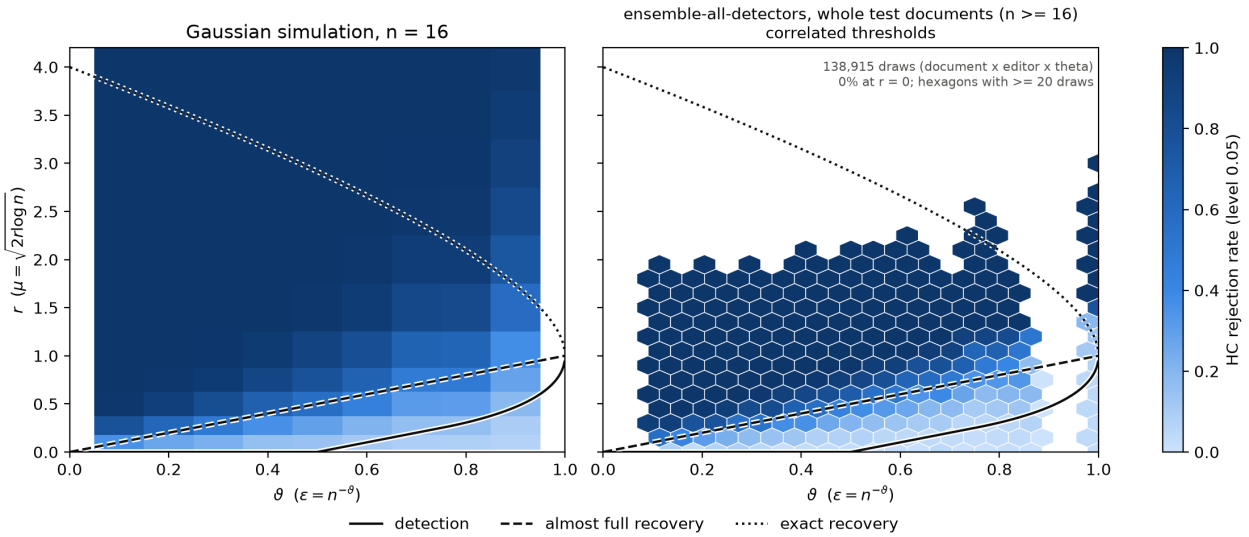


Figure 2. Left: HC power in the Gaussian rare/weak model at $n = 16$, with the boundaries for detection, almost-full recovery and exact recovery. Right: the ensemble on whole test documents, with each (document, editor, θ) triple placed at its effective signal strength r^*_{eff}. The axis labelled r in the figure is r^*, not the edit rate.

Recovery and edited fraction. Locating the edits is much harder than detecting them (Table 20). BH's Hamming error is $0.46m$ at $\theta = 0.1$ and $0.87m$ at $\theta = 0.9$. BH recovers the edited set exactly in 4–28% of documents, most often at $\theta = 0.9$, where there is only one sentence to find. Storey's estimate is close to the true edited fraction when edits are dense (0.62 vs. 0.65 at $\theta = 0.1$) and too high when they are sparse (0.19 vs. 0.04 at $\theta = 0.9$). The Meinshausen–Rice lower bound is below the true fraction in 97–100% of documents, as its 95% guarantee requires, but it is loose. At $\theta = 0.1$ it averages 0.23 while the true fraction is 0.65.

3.3 Third-party benchmarks

We also ran the ensemble through our pipeline on seven public benchmarks of AI-edited and AI-generated text: the EditLens test set [25], APT-Eval [1], OpAI-Bench [39], PASTED [40], AITDNA [41], the AI-polished peer reviews of Saha et al. [42] and a sample of RAID [43]. Each benchmark's own human texts serve as the null, per domain, so every method is compared at the same 5% false-alarm rate. The released detectors EditLens, MELD and GPTTrace v8s score each whole text, their intended use, and that score is ranked among the scores of the benchmark's human texts. The ensemble is best on the benchmarks with light or sparse edits (OpAI-Bench, APT-Eval and PASTED) and ties on the peer reviews (Table 8). It trails on AITDNA and on the EditLens test set. It flags only 35% of the fully generated RAID texts, against 77–100% for EditLens [25], MELD [44] and GPTTrace v8s [27]. MELD and GPTTrace v8s were trained on RAID [27, 44], from which our sample is drawn, so for them this benchmark is in-distribution. Under their authors' own decision thresholds the released detectors sit at different trade-offs. MELD and GPTTrace flag almost no human text but miss most light edits (MELD flags 23% of edited OpAI-Bench texts), while EditLens flags 8–26% of human texts. A null built from our own corpus does not transfer, since on OpAI-Bench it flags 18–26% of human texts for our single detectors, so the null has to come from the target domain.

4 Separating AI-edited from fully generated documents

The test of H_0 Section 2.5 (stage 1) also rejects for fully AI-generated documents, so on its own it cannot tell an AI-edited document from a generated one. We add a second test (stage 2) of H_0', that every sentence is AI-generated. To build generated documents, GPT-6 Luna wrote a one-line topic for each of the 8,446

Benchmark	Texts	Ensemble	EditLens	MELD	GPTTrace v8s
OpAI-Bench	edited	**0.70**	0.59	0.64	0.50
APT-Eval	polished	**0.58**	0.48	0.55	0.35
PASTED (test)	edited	**0.38**	0.33	0.36	0.32
Peer reviews (Saha et al.)	polished, hard	0.99	0.97	0.99	0.96
AITDNA	edited	0.87	0.90	**0.92**	0.81
EditLens (test)	edited	0.74	**0.84**	0.83	0.79
EditLens (test)	generated	0.98	1.00	1.00	1.00
RAID (train sample)	generated	0.35	0.77	1.00	1.00
All benchmarks	human (false alarm)	0.04–0.06	0.04–0.07	0.03–0.06	0.03–0.07

Table 8. Third-party benchmarks at a 5% false-alarm rate. Each cell is the share of texts flagged. The ensemble uses our pipeline (sentence p-values and HC with correlated thresholds), and EditLens, MELD and GPTTrace v8s score the whole text. Every method is calibrated against the benchmark's own human texts per domain (conformal test at $\alpha = 0.05$). MELD and GPTTrace v8s were trained on RAID.

calib and test documents. Five generators (GPT-6 Luna, gpt-oss-120b, Claude Haiku 5.5, Gemma 4 and Qwen3.5) then each wrote a document on that topic, in the same genre and of roughly the same length, without seeing the human text. Stage-2 p-values rank each sentence against generated calib sentences from the four generators other than Qwen, which is held out. They use the lower tail, so a small p-value means the sentence looks more human than generated text. Thresholds again account for within-document correlation ($\hat{\rho}' = 0.05$–0.19). The sentence detectors are not retrained for this stage. Results use a three-model ensemble of Llama-3.1-8B, OLMo-3-7B and DeBERTa-v3-large, without OLMo-3-32B, as the three model ensemble performs close enough at much lower computational cost.

- Stage 1 rejects 98.7–100% of generated test documents.
- On generated documents from the four calibration generators pooled (5,179 documents), the type I error for H_0' is 0.050 for HC, 0.045 for min-p and 0.036 for Stouffer's test. Per generator it ranges from 0.012 (GPT-6 Luna) to 0.082 (Gemma 4), so the level holds for the mixture of generators but not for each one.
- On the held-out Qwen generator, HC and min-p over-reject (0.125 and 0.131), while Stouffer's test holds the level (0.052).
- Stage 2 rejects H_0' for 99.5% of human documents and 95–99% of documents with sparse edits.
- A document is labelled human if stage 1 does not reject, fully AI if stage 1 rejects and stage 2 does not, and AI-edited if both reject (Table 9). Because stage 2 almost always rejects for edited documents, the AI-edited label is limited by stage 1. It is given to 87% of edited documents at $\theta = 0.1$ and 21% at $\theta = 0.9$.
- We also test a composite null, that at least 90% of the sentences are AI-generated, and reject it when the Meinshausen–Rice lower bound on the human fraction exceeds 0.1. Its type I error is at most 0.011, and it is rejected for 98.5% of human documents.

5 Training language models on AI-edited text

Models trained recursively on model-generated data drift from the human distribution and lose its tails [45], and AI-edited text is increasingly part of web-scale corpora. We ask whether such text helps or hurts as training data, and whether per-sentence loss weights can keep a model close to the human distribution. If that is the case, then perhaps sentence level detectors can be utilized to "clean up" new data. We first pretrain small Llama-style models from scratch with the Pythia tokenizer [46] on the train, val and calib documents of our corpus (5.8M tokens). We use two sizes, 0.3M non-embedding parameters, which is Chinchilla-optimal for 5.8M tokens [47], and 1.6M, which is larger than the data supports and which we call data-starved. Each is trained with three seeds and evaluated by perplexity on the human test documents. The training arms differ only in their data.

True document	Documents	Labelled as human	fully AI	AI-edited
Human	1,301	**0.961**	0.003	0.036
Fully AI, calibration generators	5,179	0.004	**0.949**	0.047
Fully AI, held-out generator (Qwen)	1,317	0.008	**0.870**	0.121
AI-edited, training editors, $\theta = 0.1$	13,010	0.088	0.048	**0.865**
AI-edited, training editors, $\theta = 0.5$	13,010	0.425	0.005	**0.570**
AI-edited, training editors, $\theta = 0.9$	13,010	0.788	0.003	**0.209**
AI-edited, held-out editor (Qwen), $\theta = 0.1$	2,602	0.073	0.050	**0.877**
AI-edited, held-out editor (Qwen), $\theta = 0.5$	2,602	0.450	0.004	**0.546**
AI-edited, held-out editor (Qwen), $\theta = 0.9$	2,602	0.824	0.003	**0.173**

Table 9. Three-way classification of whole test documents ($n \geq 16$) with two HC tests at $\alpha = 0.05$. Stage 1 tests H_0 (every sentence human) with p-values against human calib sentences. Stage 2 tests H_0' (every sentence AI-generated) with lower-tail p-values against fully generated calib documents from four generators. A document is labelled human if stage 1 does not reject, fully AI if stage 1 rejects and stage 2 does not, and AI-edited if both reject. Bold marks the correct label, and rows sum to 1. Share of fully generated documents labelled fully AI, by calibration generator, is 0.988 for GPT-6 Luna, 0.965 for gpt-oss-120b, 0.926 for Claude Haiku 5.5 and 0.916 for Gemma 4. Three-model ensemble (Llama-3.1-8B, OLMo-3-7B, DeBERTa-v3-large).

- **Human.** The 5.8M tokens of human text.
- **Mixture.** The same documents, but each train and val document receives edits from one randomly chosen training editor at a random θ, as in Section 2.5. Calib documents have no edits and stay human. This arm resembles real data, where some sentences are edited.
- **All-edited, 1 pass.** Every sentence of the train and val documents is replaced by one editor's edit (3.0M tokens). Its control is one pass over the human train and val documents (3.1M tokens).
- **All-edited, 10 passes.** One all-edited copy of the train and val documents for each of the 10 training editor and prompt pairs (29.7M tokens). Its controls are the human train and val documents repeated 10 times (30.8M tokens) and tokens of new human documents from the same sources, adding up to a total of 26.2M with the train and val documents.

In the mixture and all-edited arms we also try two oracle per-sentence loss weights that use the true edit labels. *Mask* gives weight 0 to edited sentences, and *strength* gives weight $1 - \min(r, 1)$.

To measure whether a model prefers AI-edited text, the paired Δ test uses 30,000 edited test sentences and their human originals. For each pair we compute the mean per-token log-probability of the edit minus that of the original, and Δ is the average over pairs. $\Delta > 0$ means the model assigns higher probability to the AI-edited version.

Arm	Train tokens	Test perplexity		Δ (nats/token)	
		Chinchilla	data-starved	Chinchilla	data-starved
human	5.8M	341.0 ± 5.0	268.6 ± 5.0	−0.122	−0.104
mixture, unweighted	5.8M	341.1 ± 6.4	266.5 ± 2.2	−0.037	−0.015
mixture, strength	5.8M	342.8 ± 6.7	266.7 ± 1.0	−0.047	−0.030
mixture, mask	5.8M	353.1 ± 5.9	280.2 ± 2.5	−0.103	−0.091
human, 1 pass (train + val)	3.1M	556.6 ± 16.4	453.0 ± 19.5	−0.145	−0.140
all-edited 1 pass, unweighted	3.0M	600.7 ± 12.5	491.9 ± 4.3	+0.042	+0.070
all-edited 1 pass, strength	3.0M	603.1 ± 14.5	504.9 ± 9.5	+0.022	+0.044
all-edited 1 pass, mask	3.0M	1265.3 ± 20.0	1113.0 ± 9.0	−0.160	−0.159
human repeated 10×	30.8M	193.5 ± 9.7	155.5 ± 0.4	−0.056	−0.025
all-edited 10 passes, unweighted	29.7M	196.5 ± 1.6	156.7 ± 1.3	+0.312	+0.382
all-edited 10 passes, strength	29.7M	199.3 ± 1.3	159.2 ± 0.5	+0.285	+0.344
all-edited 10 passes, mask	29.7M	529.7 ± 6.8	469.5 ± 2.9	−0.139	−0.124
human fresh	26.2M	161.7 ± 1.2	118.0 ± 1.3	−0.026	+0.002

Table 10. Pretraining from scratch. Test perplexity on human test documents (mean ± sd over 3 seeds) and the paired Δ test (sd over seeds 0.001–0.011). The first block matches the token count of the human corpus, the second uses one pass over the train and val documents, and the third uses about 30M tokens.

Table 10 gives the results, quoted here for the data-starved size. With the same number of tokens, AI-edited text is nearly as useful for perplexity as the human text it came from (156.7 for 10 all-edited passes vs. 155.5 for the human text repeated 10 times), and new human text is much better (118.0). The oracle weights never improve perplexity over unweighted training, and masking, which discards most tokens in the all-edited arms, is far, far worse. We then turn to the Δ test to see how style preferences are affected, and it shows the cost of AI-edited data. Training on 10 all-edited passes makes the model prefer AI-edited sentences ($\Delta = +0.38$ vs. -0.03 for the human text repeated 10 times). Strength weighting removes about 10% of that shift, and masking removes all of it at the cost of perplexity.

Because the from-scratch models are tiny and see little data, we repeated every arm as LoRA continued pretraining of released models, Pythia 410M to 6.9B [46] and OLMo-2-1B base and Instruct [48]. This gives 7 base models, 13 arms and 3 seeds, each at the base model's native context length, with the learning rate chosen by a sweep on the human arm. The picture is the same (Table 11). Before any training, every base model already prefers the edited sentence ($\Delta = 0.25$–0.33), and training on human text lowers Δ to 0.19–0.23. One pass over all-edited text raises it to 0.35–0.39 and costs perplexity (Pythia-6.9B reaches 12.52 vs. 11.99 after one pass over the human text). Strength weighting removes about 10% of that shift, and masking removes nearly all of it at a further cost in perplexity. The mixture, where only some sentences are edited, leaves perplexity unchanged (Pythia-6.9B, 11.85 vs. 11.82) but still raises Δ (0.239 vs. 0.195). Perplexity on human text alone would therefore miss this shift.

Arm	Test perplexity			Δ (nats/token)		
	Pythia-1B	Pythia-6.9B	OLMo-2-1B	Pythia-1B	Pythia-6.9B	OLMo-2-1B
base model (no training)	16.68	13.22	15.52	0.262	0.250	0.264
human	15.18	11.82	12.24	0.207	0.195	0.193
mixture, unweighted	15.23	11.85	12.27	0.245	0.239	0.239
mixture, strength	15.22	11.85	12.26	0.230	0.222	0.223
mixture, mask	15.24	11.86	12.28	0.202	0.190	0.191
human, 1 pass (train + val)	15.42	11.99	12.41	0.209	0.198	0.202
all-edited 1 pass, unweighted	16.07	12.52	12.93	0.382	0.377	0.383
all-edited 1 pass, strength	15.98	12.44	12.85	0.364	0.358	0.363
all-edited 1 pass, mask	16.64	12.98	13.39	0.224	0.215	0.211
human fresh (about 4.5× the tokens)	14.79	11.51	12.04	0.207	0.192	0.195

Table 11. LoRA continued pretraining of released models. Test perplexity on human test documents at the native context length (2048 tokens for Pythia, 4096 for OLMo-2) and the paired Δ test, mean over 3 seeds (seed sd at most 0.03 for perplexity and 0.004 for Δ). Three of the seven base models are shown, and the others agree. The 10-pass arms are omitted because they overfit at the learning rate chosen on the human arm and are being rerun.

6 Limitations and Future Work

Our edited documents are assembled from independent single-sentence edits, each made against the original document, whereas a real AI pass edits sentences together. Edit positions are drawn uniformly, while real edits are not. The null distribution is per source, so the domain must be known at test time. A null pooled over sources keeps the overall false-positive rate but is miscalibrated within sources (human FineWeb-Edu documents are flagged 13–17% of the time, Table 23). Equicorrelation is the simplest dependence model. Under it, HC^+ of GPTTrace v8s and min-p of OLMo-3-32B still reach type I errors of 0.063 and 0.071, against 0.053 and 0.050 with thresholds calibrated on calib documents, which need no dependence model. Released detectors are applied to single sentences, which they were not trained for, so we also report their whole-document results. The from-scratch pretraining models are very small. In every pretraining run so far, the per-sentence weights shift emphasis between tokens rather than reducing how much the model learns from AI-edited text. Our detectors are trained on edits only and detect fully generated text from some other sources poorly (35% on RAID, while staying at 98% on EditLens), and on third-party data the null must come from the target domain.

- **A second held-out editor.** Claude Sonnet 5.5 [49] edited 20,000 sentences (6,265 of them test) under both prompts, but was not included in this study due to time. Scoring them tests whether the detectors generalize to an unseen editor from the same family as a training editor (Claude Haiku 5.5). We also plan to fine-tune the detectors on these edits.
- **Document-context detectors.** Detectors that score every sentence of a document in one pass are not yet trained. With these detectors a sentence's score depends on its neighbors, so the global null must be calibrated on calib documents rather than calib sentences.
- **Correlation beyond equicorrelation.** PC adjustment with more than one component, and innovated HC with a covariance estimated from windows of calib documents (a Toeplitz model, for correlation between neighboring sentences).
- **Other released detectors.** ImBD [50] and DetectAnyLLM [51] are candidates.

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A Detector training details

Search. We searched 29 configurations on 20% of the training documents for 2 epochs (Table 13). They cover three families. The first is frozen embedding models with a logistic-regression head (Qwen3-Embedding [52], EmbeddingGemma 2 [53], Llama-Embed-Nemotron [54]). The second is fully fine-tuned encoders (DeBERTa-v3-large [24], ModernBERT-large, Ettin-encoder-1B [55]). The third is decoders with LoRA [21] and a classification head on the last token (Llama-3.1-8B [22], OLMo-3-7B [23], Ettin-decoder-1B). Models were selected by TPR at 1% FPR on the training editors' val sentences (12,335 human, 110,998 edited). Fine-tuned models reached 0.30–0.32, and frozen embedding models reached at most 0.17. In every fine-tuned family the lowest learning rate tried won in each family except for ModernBERT-large.

Full training. The three best families were trained on all training documents for three epochs, with learning rates at the search winner and a factor of two on either side. We evaluated on val ten times per epoch and kept the best checkpoint (Table 12). OLMo-3-32B [23] was added later and trained for one epoch only, because of the compute time available. Decoders use LoRA with rank 16, $\alpha = 32$ and dropout 0.05 on all linear layers. Inputs are capped at 256 tokens.

Backbone	Learning rates tried (TPR@1%)	Chosen	AUROC	TPR@0.1%	TPR@1%	TPR@5%
Llama-3.1-8B (LoRA)	2.5e-5 / 5e-5 / 1e-4 (0.351 / 0.350 / 0.326)	2.5e-5	0.917	0.175	0.351	0.572
OLMo-3-7B (LoRA)	2.5e-5 / 5e-5 / 1e-4 (0.349 / 0.349 / 0.346)	2.5e-5	0.912	0.181	0.349	0.567
DeBERTa-v3-large	2.5e-6 / 5e-6 / 1e-5 (0.337 / 0.342 / 0.334)	5e-6	0.916	0.169	0.342	0.570
OLMo-3-32B (LoRA, 1 epoch)	5e-5	5e-5	0.923	0.178	0.374	0.609

Table 12. Full training runs. Val TPR at 1% FPR for each learning rate tried, and val metrics of the chosen run, on the training editors' val sentences. The held-out editor is never used for selection. The best OLMo-3-7B checkpoint came in epoch 2. Full training adds about 3 points over the search runs (0.30–0.32).

Run (model, lr, class weighting)	AUROC	TPR@0.1%	TPR@1%	TPR@5%
llama-3.1-8b_lr5e-05	0.908	0.126	0.321	0.547
olmo-3-7b_lr5e-05	0.899	0.149	0.320	0.520
deberta-v3-large_lr5e-06_per_sentence	0.902	0.130	0.301	0.522
deberta-v3-large_lr5e-06_global	0.905	0.128	0.299	0.520
deberta-v3-large_lr1e-05_per_sentence	0.902	0.128	0.298	0.522
deberta-v3-large_lr2e-05_global	0.894	0.112	0.287	0.493
deberta-v3-large_lr2e-05_per_sentence	0.894	0.105	0.287	0.493
olmo-3-7b_lr0.0002	0.891	0.140	0.286	0.492
deberta-v3-large_lr1e-05_global	0.902	0.129	0.282	0.509
ettin-encoder-1b_lr1e-05_global	0.900	0.135	0.279	0.509
ettin-encoder-1b_lr1e-05_per_sentence	0.898	0.139	0.278	0.500
ettin-encoder-1b_lr2e-05_global	0.894	0.130	0.277	0.492
ettin-decoder-1b_lr5e-05	0.889	0.127	0.274	0.489
ettin-encoder-1b_lr2e-05_per_sentence	0.892	0.127	0.271	0.490
modernbert-large_lr3e-05_global	0.887	0.120	0.268	0.469
modernbert-large_lr1e-05_global	0.892	0.106	0.264	0.481
ettin-decoder-1b_lr0.0002	0.889	0.133	0.263	0.491
modernbert-large_lr1e-05_per_sentence	0.890	0.112	0.261	0.478
modernbert-large_lr3e-05_per_sentence	0.887	0.116	0.259	0.471
modernbert-large_lr5e-05_global	0.883	0.115	0.249	0.466
modernbert-large_lr5e-05_per_sentence	0.877	0.111	0.247	0.451
ettin-encoder-1b_lr4e-05_global	0.878	0.110	0.244	0.451
ettin-encoder-1b_lr4e-05_per_sentence	0.877	0.114	0.238	0.451
llama-3.1-8b_lr0.0002	0.851	0.087	0.187	0.396
llama-embed-nemotron-8b	0.832	0.075	0.174	0.363
qwen3-emb-8b	0.827	0.041	0.140	0.347
qwen3-emb-0.6b	0.784	0.056	0.121	0.277
qwen3-emb-4b	0.793	0.017	0.094	0.265
embeddinggemma-2	0.710	0.012	0.059	0.172

Table 13. Detector search on 20% of training documents, 2 epochs, val metrics on the training editors. Run names give the model, the learning rate and the class weighting. Global weighting uses inverse class frequency. Per-sentence weighting gives each human sentence and the set of its edits the same total weight.

B Released detector setup

EditLens is set up as in Pangram's release, with a 4-bit NF4 base model [56], bf16 compute, Pangram's text cleaning and a 1024-token limit. On 600 rows of Pangram's own test set, our scores match their published ones with Pearson correlation 0.99999 and mean absolute difference 0.0008, and the predicted bucket agrees on 597 rows, strongly suggesting that our implementation is correct and results are representative. GPTTrace Ettin-150M is run as the released int8 ONNX model with its homoglyph normalization. desklib and GPTTrace v8s are also used as released.

C Correlation model and thresholds

For each source and detector, we estimate ρ by the one-way ANOVA intraclass correlation of z on calib documents. The estimate is cross-fitted. Calib documents are split into two folds, and the p-values of each fold are computed against the other fold. For each test and each n, the critical value is the 95% quantile of the statistic over 100,000 Monte Carlo draws from the equicorrelated null with $\rho = \hat{\rho}$.

D Thresholds from calib documents and decorrelated HC

Calib-document thresholds. As a model-free alternative, i.e without first modeling the calibration set then drawing from it, we run every global test as a document-level conformal test. The test document's statistic is ranked among the same statistic on calib human documents of the same source and length band, with cross-fitted p-values (79–491 calib documents per source and at least 60 per band). This holds level α for any dependence within documents, provided calib and test human documents are exchangeable. For HC it changes almost nothing (type I error 0.043–0.046, power within 0.03, Tables 21 and 22).

Decorrelated HC. PC-adjusted HC (one component) and innovated HC [57] remove the correlation before testing. Under equicorrelation the top principal component is the all-ones vector, so PC adjustment subtracts the document's mean z. The innovated transform reduces to $z_j - c\bar{z}$ with $c = n\hat{\rho}/(1+(n-1)\hat{\rho})$. With Monte Carlo thresholds both variants over-reject (type I error 0.08–0.10), because the rescaling assumes that the within-document spread of every document is $1-\hat{\rho}$. With calib-document thresholds they hold their level (0.04–0.06). At matched size they beat HC only when edits are sparsest (ensemble at $\theta = 0.9$, 0.28 vs. 0.21 for HC and 0.26 for min-p) and lose heavily when edits are dense (0.23 and 0.44 vs. 0.91 at $\theta = 0.1$), because centering also subtracts the edits' own mean.

E Pooled null

Here each sentence is ranked against the calib sentences of all sources, still within length bins. Sentence-level TPR is nearly unchanged and the overall FPR stays at 1% (Table 23). The FPR per source, however, spreads to 0.004–0.022 (0.007–0.015 with the per-source null). At the document level, human FineWeb-Edu documents are flagged 13–17% of the time and human XSum documents never. With a pooled $\hat{\rho}$ (0.30–0.33, which absorbs differences between sources), the overall type I error is 0.036–0.040 and HC power drops by 1–5 points.

F Additional tables

	Ensemble		OLMo-3-32B		EditLens (3B)	
Source	train	Qwen	train	Qwen	train	Qwen
FineWeb-Edu	0.222	0.193	0.218	0.193	0.052	0.067
peS2o	0.434	0.375	0.432	0.382	0.046	0.058
PERSUADE 2.0	0.442	0.428	0.438	0.419	0.109	0.121
ELLIPSE	0.611	0.600	0.618	0.607	0.231	0.229
XSum	0.368	0.351	0.353	0.333	0.051	0.068
OpenReview	0.415	0.427	0.398	0.410	0.081	0.116
Enron	0.334	0.288	0.303	0.257	0.054	0.055
WritingPrompts	0.243	0.200	0.228	0.183	0.058	0.056

Table 14. TPR at 1% FPR by source, test split.

	i.i.d. thresholds		Equicorrelated thresholds				
Detector	HC	min-p	HC	HC$^+$	min-p	Stouffer	$\hat{\rho}$ (by source)
Ensemble	0.085	0.046	0.045	0.038	0.052	0.042	0.07–0.32
OLMo-3-32B	0.095	0.058	0.054	0.040	0.071	0.038	0.06–0.32
Llama-3.1-8B	0.083	0.043	0.048	0.048	0.049	0.042	0.06–0.30
OLMo-3-7B	0.088	0.047	0.048	0.048	0.054	0.044	0.07–0.31
DeBERTa-v3-large	0.088	0.048	0.041	0.038	0.053	0.041	0.06–0.29
EditLens (3B)	0.104	0.053	0.058	0.051	0.056	0.041	0.08–0.28
GPTTrace v8s (1B)	0.069	0.036	0.049	0.063	0.040	0.050	0.06–0.13
desklib v1.01	0.066	0.048	0.053	0.057	0.049	0.049	0.04–0.12
GPTTrace Ettin-150M	0.084	0.047	0.058	0.058	0.048	0.058	0.04–0.16

Table 15. Type I error on the 1,301 human test documents at $\alpha = 0.05$, with thresholds for independent sentences and for equicorrelated sentences, and the range of $\hat{\rho}$ over sources.

Detector	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
Ensemble	0.92/0.93	0.78/0.78	0.59/0.56	0.41/0.38	0.22/0.19
OLMo-3-32B	0.91/0.93	0.77/0.77	0.58/0.55	0.41/0.38	0.22/0.19
Llama-3.1-8B	0.89/0.89	0.72/0.69	0.53/0.47	0.37/0.30	0.20/0.16
OLMo-3-7B	0.89/0.91	0.72/0.72	0.52/0.49	0.36/0.31	0.19/0.16
DeBERTa-v3-large	0.90/0.92	0.75/0.76	0.55/0.53	0.38/0.35	0.21/0.18
EditLens (3B)	0.46/0.50	0.28/0.31	0.17/0.18	0.12/0.14	0.08/0.08
GPTTrace v8s (1B)	0.35/0.29	0.26/0.17	0.19/0.11	0.14/0.08	0.09/0.06
desklib v1.01	0.14/0.14	0.09/0.10	0.07/0.08	0.06/0.06	0.06/0.06
GPTTrace Ettin-150M	0.13/0.14	0.09/0.11	0.07/0.08	0.07/0.07	0.06/0.06
EditLens (3B) (whole doc.)	0.53/0.56	0.25/0.23	0.13/0.11	0.08/0.07	0.06/0.05
GPTTrace v8s (1B) (whole doc.)	0.39/0.26	0.21/0.10	0.13/0.07	0.10/0.06	0.07/0.06
desklib v1.01 (whole doc.)	0.35/0.34	0.18/0.15	0.11/0.09	0.08/0.06	0.06/0.05
GPTTrace Ettin-150M (whole doc.)	0.09/0.08	0.06/0.06	0.05/0.05	0.05/0.05	0.05/0.05

Table 16. HC rejection rate on edited whole test documents ($n \geq 16$) with equicorrelated thresholds. Each cell shows training editors / held-out Qwen editor. The last block scores each whole document with the released detector directly (document-level conformal test against calib human documents, type I error 0.050–0.055).

Detector	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
Ensemble	0.83/0.84	0.73/0.72	0.59/0.56	0.45/0.42	0.27/0.23
OLMo-3-32B	0.83/0.84	0.72/0.72	0.59/0.56	0.46/0.42	0.28/0.25
Llama-3.1-8B	0.78/0.75	0.67/0.62	0.54/0.48	0.41/0.34	0.24/0.19
OLMo-3-7B	0.79/0.79	0.68/0.64	0.53/0.49	0.39/0.35	0.23/0.19
DeBERTa-v3-large	0.81/0.82	0.70/0.70	0.56/0.54	0.42/0.39	0.25/0.21
EditLens (3B)	0.35/0.41	0.25/0.29	0.17/0.20	0.13/0.15	0.09/0.09

Table 17. Min-p rejection rate on whole test documents, training editors / held-out Qwen editor.

Detector	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
Ensemble	0.87/0.88	0.45/0.43	0.18/0.17	0.10/0.09	0.06/0.06
OLMo-3-32B	0.87/0.89	0.45/0.44	0.18/0.18	0.09/0.09	0.06/0.05
Llama-3.1-8B	0.86/0.85	0.43/0.40	0.17/0.16	0.09/0.08	0.06/0.05
OLMo-3-7B	0.85/0.86	0.41/0.40	0.17/0.17	0.09/0.09	0.06/0.06
DeBERTa-v3-large	0.87/0.88	0.44/0.44	0.18/0.18	0.10/0.09	0.06/0.06
EditLens (3B)	0.38/0.40	0.16/0.16	0.08/0.09	0.06/0.06	0.05/0.05

Table 18. Stouffer rejection rate on whole test documents, training editors / held-out Qwen editor.

Detector	n	Windows	Type I	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
Ensemble	16	1,301	0.053	0.89	0.77	0.61	0.42	0.27
	32	443	0.052	0.90	0.72	0.54	0.37	0.18
OLMo-3-32B	16	1,301	0.052	0.88	0.76	0.60	0.42	0.26
	32	443	0.056	0.89	0.71	0.54	0.36	0.18
Llama-3.1-8B	16	1,301	0.045	0.85	0.72	0.55	0.37	0.24
	32	443	0.056	0.86	0.65	0.47	0.32	0.16
OLMo-3-7B	16	1,301	0.054	0.85	0.72	0.55	0.36	0.23
	32	443	0.045	0.87	0.67	0.49	0.31	0.16
DeBERTa-v3-large	16	1,301	0.049	0.87	0.74	0.58	0.39	0.25
	32	443	0.045	0.88	0.68	0.50	0.34	0.17

Table 19. Fixed-length windows. For each test document with at least n sentences, one random window of n consecutive sentences is tested. HC type I error and power with equicorrelated thresholds.

θ	$\bar{m}$	ε	$\hat{\varepsilon}_{\text{Storey}}$	$\hat{\varepsilon}_{\text{MR}}$	MR covers	BH Hamming/m	BH exact	Bonf. Hamming/m	Bonf. exact
0.1	22.51	0.65	0.62	0.23	1.00	0.46	0.05	0.78	0.00
0.2	15.48	0.47	0.46	0.14	1.00	0.53	0.04	0.78	0.00
0.3	10.70	0.34	0.35	0.09	1.00	0.59	0.05	0.78	0.01
0.4	7.38	0.24	0.29	0.05	0.99	0.64	0.06	0.78	0.01
0.5	5.11	0.17	0.25	0.03	0.99	0.68	0.07	0.79	0.02
0.6	3.61	0.12	0.22	0.02	0.99	0.73	0.09	0.79	0.04
0.7	2.54	0.09	0.21	0.02	0.98	0.77	0.12	0.79	0.08
0.8	1.86	0.07	0.20	0.01	0.98	0.81	0.16	0.79	0.11
0.9	1.00	0.04	0.19	0.01	0.97	0.87	0.28	0.77	0.26

Table 20. Ensemble on whole test documents. Mean number of edited sentences, true and estimated edited fraction, Meinshausen–Rice coverage, and support recovery by BH ($q = 0.1$) and Bonferroni ($\alpha = 0.05$).

		Type I		Power, calib-document thresholds				
Detector	Test	MC	calib docs	$\theta = 0.1$	$\theta = 0.3$	$\theta = 0.5$	$\theta = 0.7$	$\theta = 0.9$
Ensemble	HC	0.045	0.046	0.91	0.77	0.58	0.40	0.21
	HC^+	0.038	0.034	0.53	0.52	0.17	0.08	0.05
	min-p	0.052	0.053	0.83	0.72	0.58	0.44	0.26
	Stouffer	0.042	0.038	0.86	0.43	0.18	0.09	0.05
	PC-adjusted HC	0.101	0.064	0.23	0.52	0.51	0.42	0.28
	innovated HC	0.097	0.057	0.44	0.61	0.55	0.45	0.28
OLMo-3-32B	HC	0.054	0.046	0.90	0.76	0.57	0.40	0.21
	HC^+	0.040	0.038	0.54	0.53	0.18	0.08	0.05
	min-p	0.071	0.050	0.81	0.69	0.56	0.42	0.25
	Stouffer	0.038	0.038	0.86	0.44	0.18	0.10	0.05
	PC-adjusted HC	0.105	0.058	0.22	0.49	0.49	0.41	0.27
	innovated HC	0.100	0.059	0.43	0.59	0.54	0.44	0.28
Llama-3.1-8B	HC	0.048	0.044	0.87	0.71	0.51	0.35	0.19
	HC^+	0.048	0.038	0.56	0.49	0.16	0.08	0.05
	min-p	0.049	0.048	0.76	0.64	0.51	0.38	0.22
	Stouffer	0.042	0.040	0.84	0.40	0.16	0.08	0.05
	PC-adjusted HC	0.093	0.051	0.19	0.43	0.43	0.36	0.23
	innovated HC	0.090	0.049	0.37	0.52	0.47	0.38	0.23
OLMo-3-7B	HC	0.048	0.043	0.87	0.70	0.50	0.33	0.17
	HC^+	0.048	0.039	0.59	0.51	0.18	0.08	0.05
	min-p	0.054	0.042	0.75	0.63	0.48	0.35	0.19
	Stouffer	0.044	0.046	0.85	0.42	0.18	0.10	0.06
	PC-adjusted HC	0.090	0.048	0.17	0.41	0.41	0.34	0.21
	innovated HC	0.088	0.044	0.33	0.50	0.45	0.36	0.22
DeBERTa-v3-large	HC	0.041	0.043	0.90	0.74	0.55	0.37	0.20
	HC^+	0.038	0.047	0.58	0.56	0.22	0.10	0.06
	min-p	0.053	0.043	0.80	0.68	0.54	0.40	0.23
	Stouffer	0.041	0.047	0.86	0.44	0.18	0.10	0.06
	PC-adjusted HC	0.082	0.042	0.18	0.44	0.44	0.36	0.23
	innovated HC	0.078	0.042	0.37	0.54	0.48	0.38	0.23

Table 21. Whole test documents, per-source null, all editors pooled. Every global test with thresholds from calib documents (the statistic ranked among calib human documents of the same source and length band). MC is the type I error with the Monte Carlo thresholds of the equicorrelated model, for comparison. PC-adjusted (one component) and innovated HC over-reject with the Monte Carlo thresholds and hold their level with the calib-document thresholds.

Detector	Type I	$\theta = 0.1$	$\theta = 0.5$	$\theta = 0.9$
Ensemble	0.046	0.91/0.93	0.58/0.56	0.22/0.19
OLMo-3-32B	0.046	0.90/0.92	0.57/0.55	0.22/0.19
Llama-3.1-8B	0.044	0.87/0.87	0.52/0.45	0.19/0.14
OLMo-3-7B	0.043	0.87/0.89	0.50/0.46	0.17/0.14
DeBERTa-v3-large	0.043	0.89/0.92	0.55/0.53	0.21/0.18
EditLens (3B)	0.050	0.42/0.46	0.15/0.16	0.07/0.07
GPTTrace v8s (1B)	0.046	0.34/0.27	0.18/0.10	0.09/0.05
desklib v1.01	0.041	0.11/0.12	0.06/0.06	0.04/0.04
GPTTrace Ettin-150M	0.048	0.12/0.13	0.07/0.07	0.05/0.05
Fast-DetectGPT (Llama-3-8B)	0.063	0.12/0.10	0.08/0.07	0.07/0.07
Fast-DetectGPT (Falcon-7B)	0.062	0.10/0.07	0.07/0.06	0.06/0.06
Log-likelihood (Llama-3-8B)	0.039	0.11/0.08	0.05/0.05	0.04/0.04
Log-likelihood (Falcon-7B)	0.042	0.11/0.08	0.06/0.05	0.04/0.04
Binoculars (Llama-3-8B)	0.051	0.12/0.10	0.07/0.06	0.06/0.05
Binoculars (Falcon-7B)	0.051	0.11/0.08	0.06/0.06	0.05/0.05

Table 22. HC with calib-document thresholds for every detector (compare the detection columns of Table 6). Each cell shows training editors / held-out Qwen editor.

			Sentences, $p < 0.01$				Whole documents, HC				
Detector	Null	$\hat{\rho}$	train	Qwen	FPR	FPR by source	$\hat{\rho}$ range	Type I	$\theta = 0.1$	$\theta = 0.5$	$\theta = 0.9$
Ensemble	per source	per source	0.386	0.362	0.0105	0.0071–0.0148	0.07–0.32	0.045	0.92	0.59	0.22
	pooled	pooled	0.383	0.353	0.0101	0.0036–0.0217	0.33–0.33	0.036	0.89	0.56	0.21
	pooled	per source	0.383	0.353	0.0101	0.0036–0.0217	0.06–0.29	0.053	0.93	0.60	0.24
OLMo-3-32B	per source	per source	0.377	0.352	0.0111	0.0081–0.0154	0.06–0.32	0.054	0.91	0.58	0.22
	pooled	pooled	0.376	0.347	0.0109	0.0048–0.0212	0.32–0.32	0.038	0.89	0.55	0.21
	pooled	per source	0.376	0.347	0.0109	0.0048–0.0212	0.06–0.29	0.055	0.93	0.60	0.24
Llama-3.1-8B	per source	per source	0.341	0.295	0.0102	0.0085–0.0136	0.06–0.30	0.048	0.89	0.52	0.19
	pooled	pooled	0.334	0.282	0.0101	0.0023–0.0268	0.32–0.32	0.036	0.84	0.48	0.18
	pooled	per source	0.334	0.282	0.0101	0.0023–0.0268	0.06–0.27	0.049	0.90	0.53	0.20
OLMo-3-7B	per source	per source	0.331	0.302	0.0102	0.0067–0.0130	0.07–0.31	0.048	0.89	0.52	0.18
	pooled	pooled	0.334	0.302	0.0099	0.0041–0.0191	0.30–0.30	0.040	0.86	0.50	0.18
	pooled	per source	0.334	0.302	0.0099	0.0041–0.0191	0.06–0.28	0.051	0.90	0.54	0.20
DeBERTa-v3-large	per source	per source	0.341	0.316	0.0101	0.0071–0.0150	0.06–0.29	0.041	0.91	0.55	0.20
	pooled	pooled	0.334	0.307	0.0101	0.0035–0.0223	0.31–0.31	0.037	0.87	0.50	0.18
	pooled	per source	0.334	0.307	0.0101	0.0035–0.0223	0.06–0.27	0.051	0.91	0.54	0.20

Table 23. Per-source vs. pooled conformal null (10 length bins in both). For sentences, TPR at $p < 0.01$ for training editors and the held-out Qwen editor, the overall FPR on human test sentences, and its range over the 8 sources. For whole documents, the range of $\hat{\rho}$ over sources, and HC type I error and power (all editors pooled). Rows with a pooled null and per-source $\hat{\rho}$ use the domain label only for $\hat{\rho}$.